

INTEGUMENTARY SYSTEM

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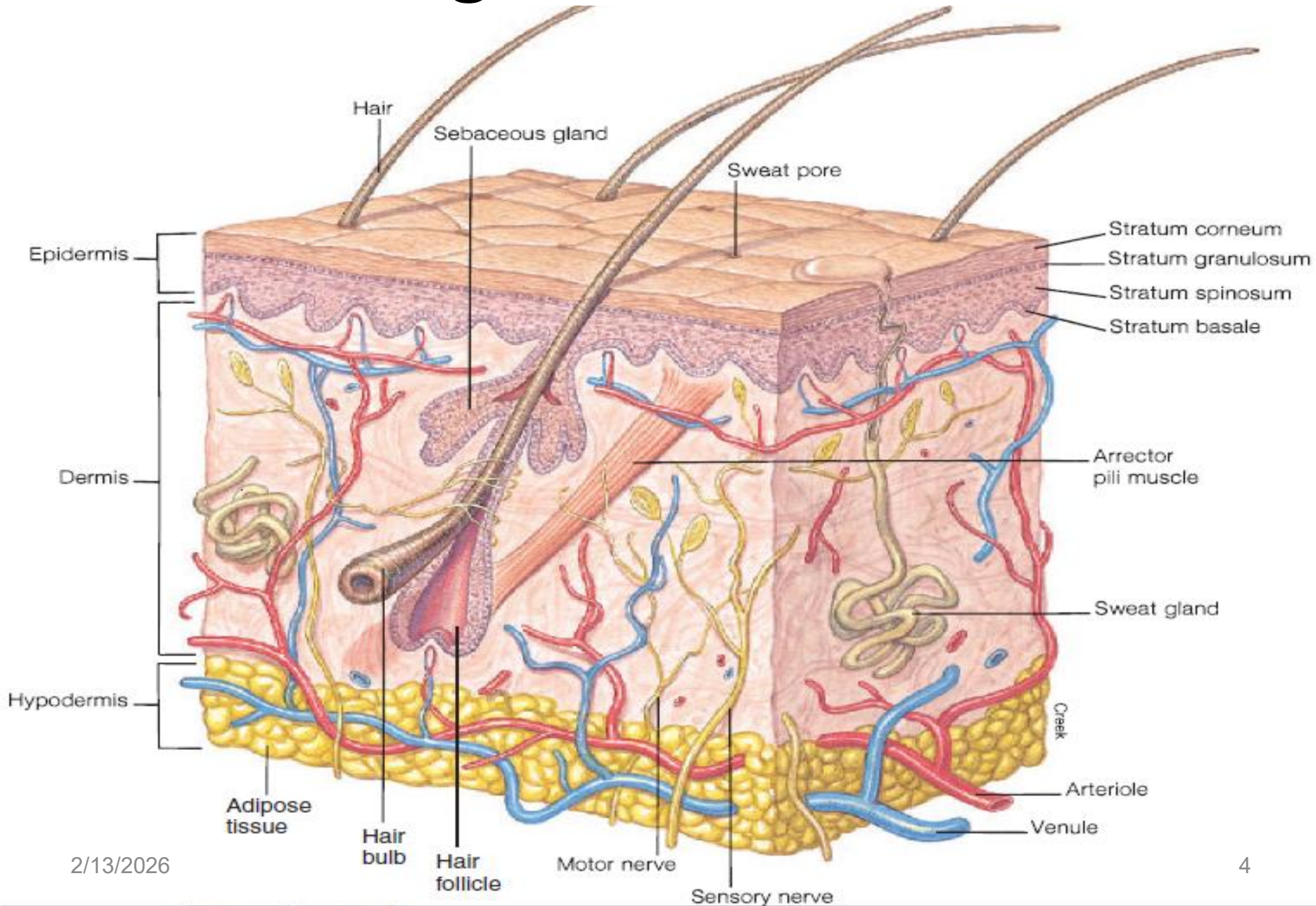
Constitutes of

- skin
- And its accessory organs (hair, glands, and nails).
- Skin is the largest organ of the body
- Interface between the continually changing external environment and the body's internal environment

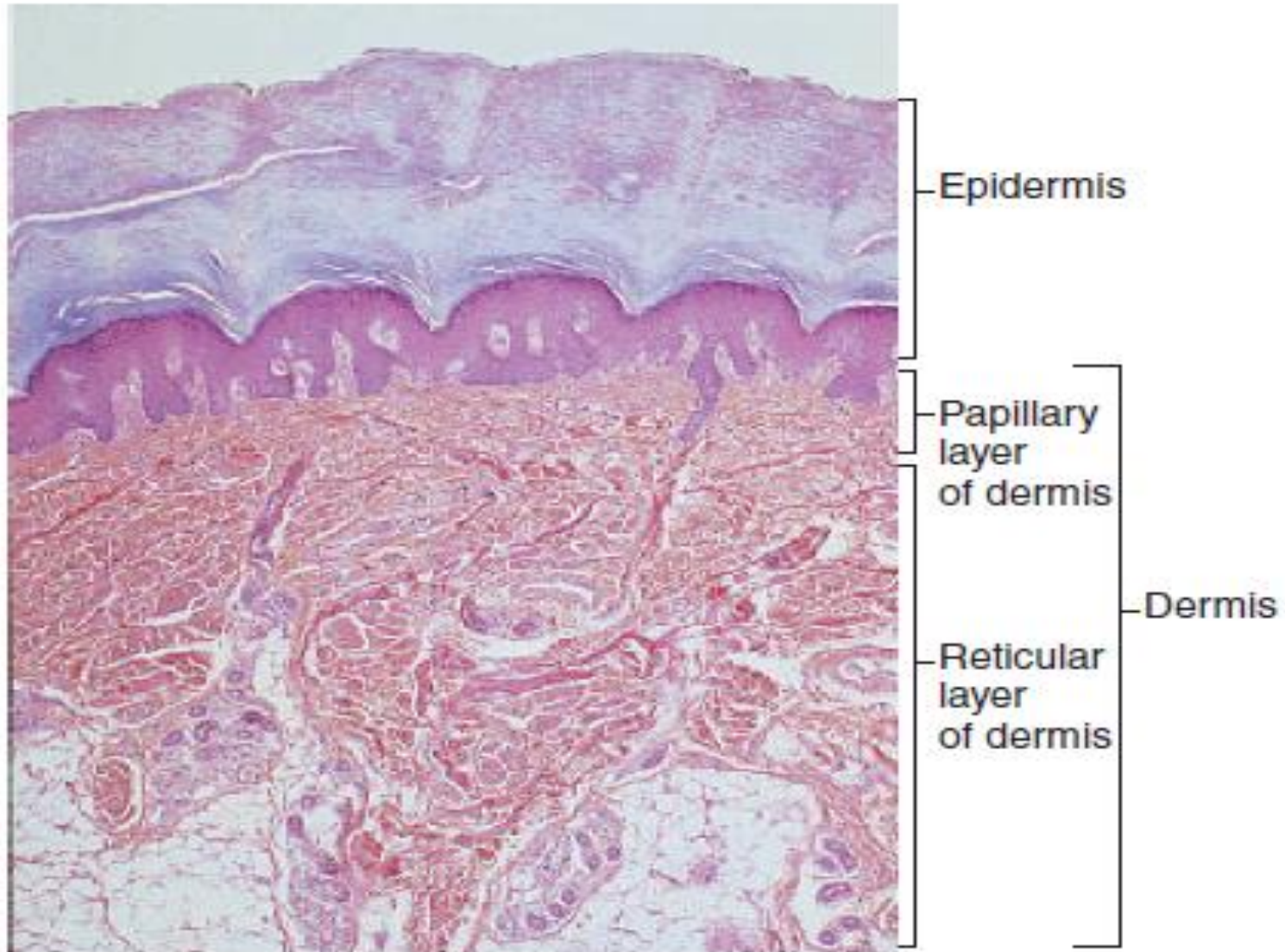
Functions of the skin

- Physical Protection
- Hydroregulation
- Thermoregulation
- Cutaneous Absorption
- Synthesis
- Sensory Reception
- Communication
- Excretory

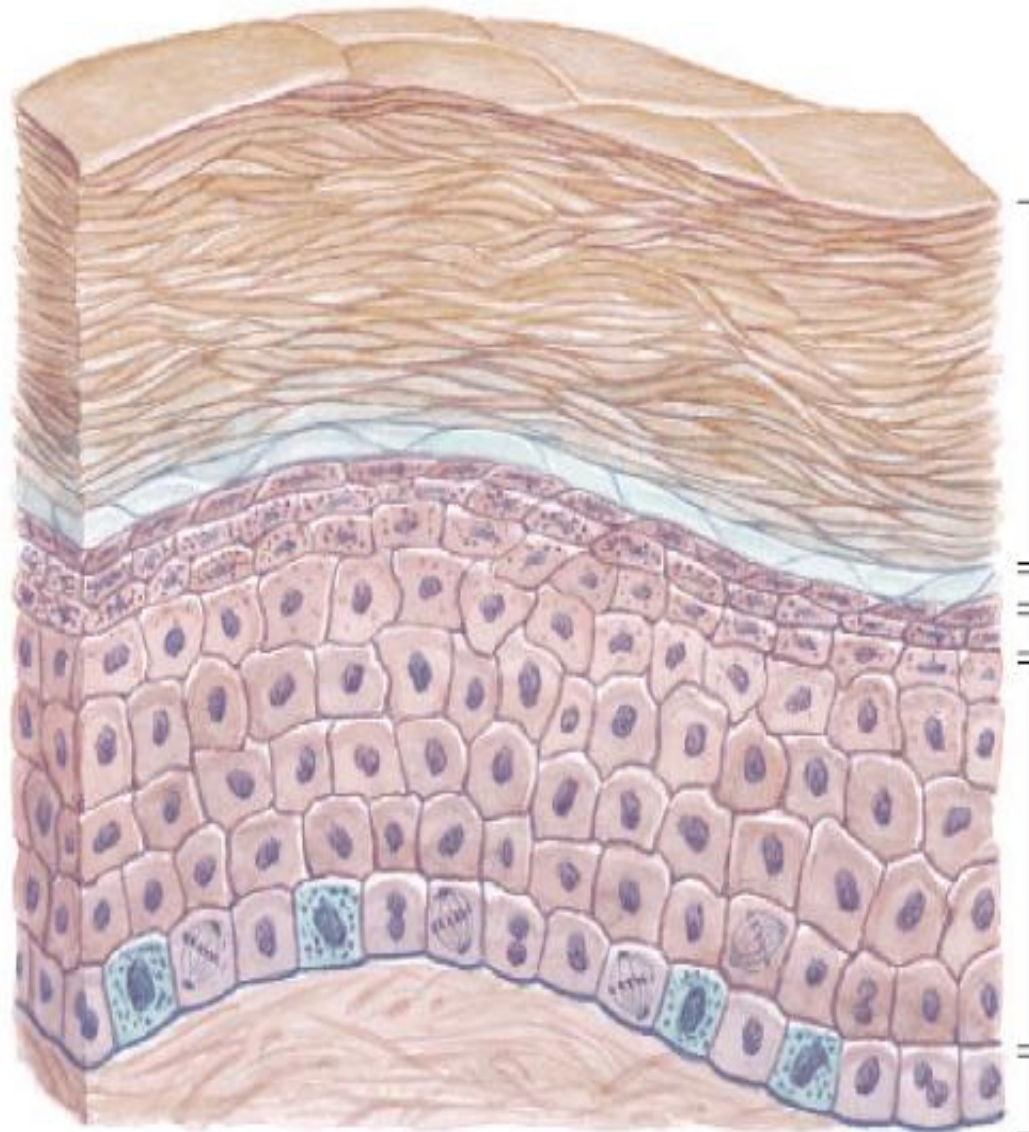
A diagram of the skin.



Layers of the skin



Epidermal layers



Stratum corneum

Consists of many layers of keratinized dead cells that are flattened and nonnucleated; cornified

Stratum lucidum

A thin, clear layer found only in the epidermis of the lips, palms, and soles

Stratum granulosum

Composed of one or more layers of granular cells that contain fibers of keratin and shriveled nuclei

Stratum spinosum

Composed of several layers of cells with centrally located, large, oval nuclei and spinelike processes; limited mitosis

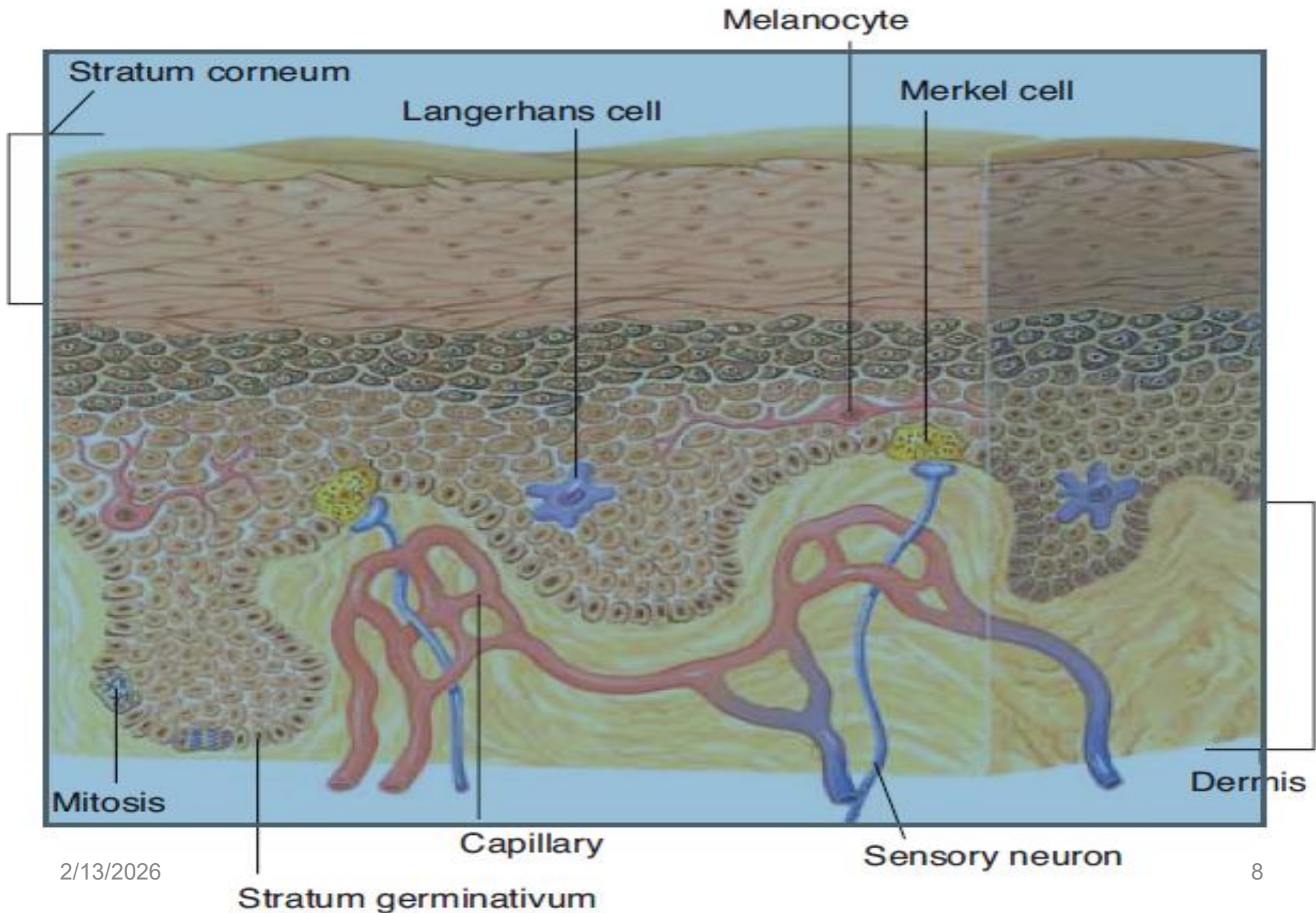
Stratum basale

Consists of a single layer of cuboidal cells in contact with the basement membrane that undergo mitosis; contains pigment-producing melanocytes

Cells of the epidermis

- **Keratinocytes:** Keratinocytes are the most abundant cells in the epidermis and play a central role in maintaining the skin's barrier function. These cells produce the protein keratin, which provides strength, structure, and waterproofing to the skin. As keratinocytes move upward through the layers of the epidermis, they undergo the process of cornification, where they become flattened, lose their nuclei, and form the protective outermost layer of the skin known as the **stratum corneum**.
- **Melanocytes:** Melanocytes are specialized cells responsible for producing the pigment melanin. Melanin provides color to the skin, hair, and eyes and helps protect the skin from the harmful effects of ultraviolet (UV) radiation from the sun. Melanocytes transfer melanin to nearby keratinocytes, which offers some level of protection against DNA damage caused by UV exposure.
- **Tactile Cells (Merkel Cells):** Tactile cells, also known as Merkel cells, are found in the epidermal-dermal junction and are involved in the sense of touch. They are associated with nerve endings and play a role in detecting pressure and touch sensations. Tactile cells are particularly concentrated in areas of the skin that are highly sensitive, such as the fingertips.
- **Nonpigmented Granular Dendrocytes (Langerhans Cells):** Langerhans cells are a type of immune cell found in the epidermis. They are specialized antigen-presenting cells that play a crucial role in the body's immune response. Langerhans cells capture antigens (foreign substances) that come into contact with the skin and then migrate to nearby lymph nodes, where they present these antigens to other immune cells, initiating an immune response.

Cells of the epidermis



Coloration of the Skin

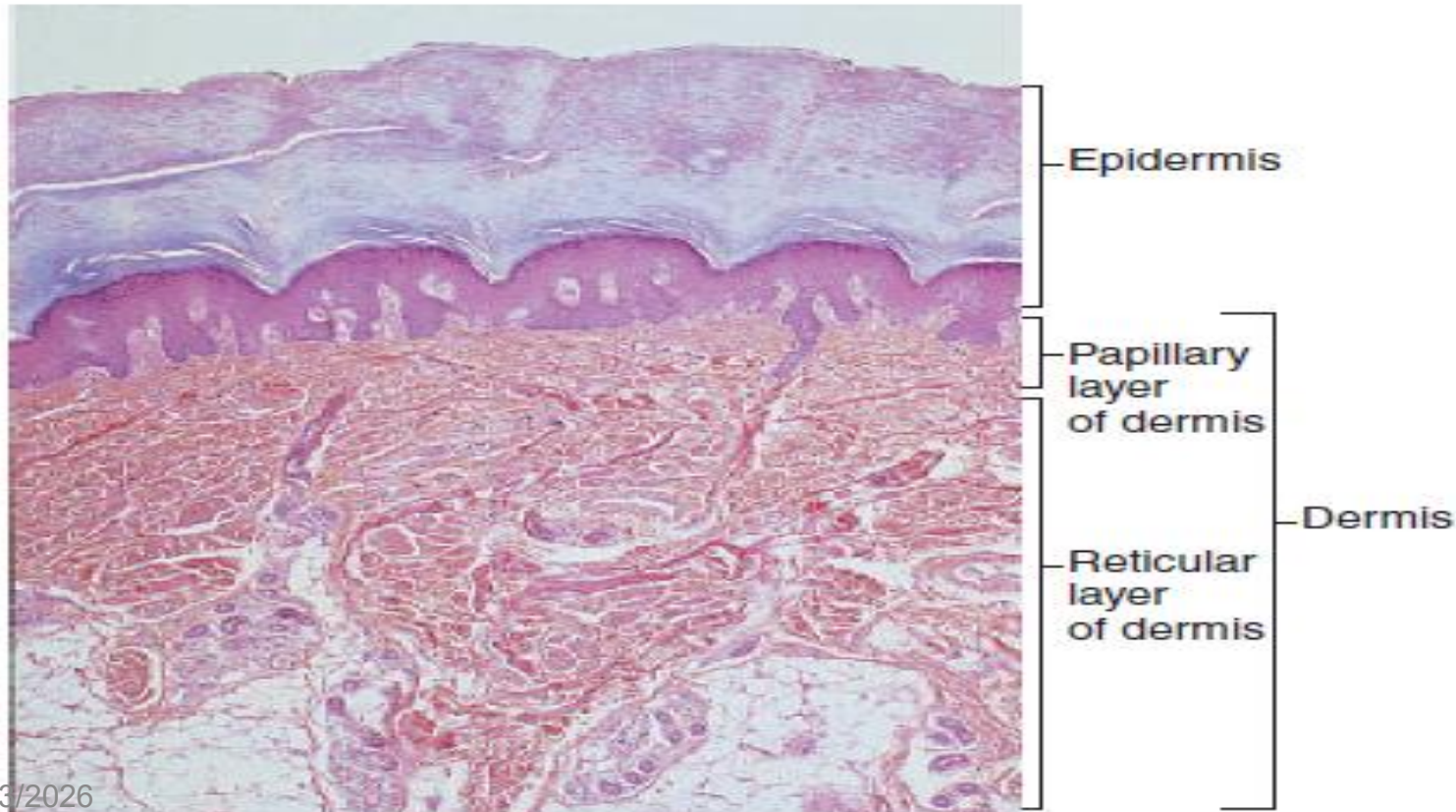
- **Melanin:** Melanin is a pigment responsible for the color of the skin, hair, and eyes. It is produced by melanocytes, specialized cells located in the epidermis. The amount and type of melanin present in the skin determine an individual's skin color. Melanin serves to protect the skin from the harmful effects of ultraviolet (UV) radiation by absorbing and dissipating UV energy. People with more melanin are generally more protected from the sun's damaging rays, while those with less melanin are more susceptible to sunburn and other UV-related skin damage.
- **The Level of Oxygenation of Hemoglobin:** The color of the skin can also be influenced by the level of oxygenation of hemoglobin, the protein in red blood cells that carries oxygen. When hemoglobin is oxygenated, it appears bright red and contributes to the pinkish hue of the skin, particularly in areas where blood vessels are close to the surface, such as the cheeks. Deoxygenated hemoglobin has a bluish hue and contributes to the color of veins seen through the skin.
- **The Amount of Blood Circulating in the Dermis:** The amount of blood circulating in the dermis, the deeper layer of the skin, can affect skin color. When blood vessels in the dermis dilate (expand), such as during increased blood flow due to exercise or increased body temperature, the skin may appear redder. Conversely, constriction of blood vessels can lead to a paler appearance.
- **Bile Pigments in Blood:** Bile pigments, such as bilirubin, can contribute to skin coloration. An accumulation of bilirubin in the bloodstream due to liver dysfunction or other health conditions can lead to a yellowish discoloration of the skin and eyes, a condition known as jaundice.
- **Carotenes:** Carotenes are orange-yellow pigments found in certain foods, particularly fruits and vegetables. They can accumulate in the skin when consumed in large amounts. For example, a diet rich in beta-carotene, a type of carotene found in carrots and other colorful vegetables, can lead to a slightly orange tint to the skin. This condition is called carotenemia and is generally harmless.

Dermis

- The dermis is deeper and thicker than the epidermis

Layers of the Dermis

- Stratum papillarosum (papillary layer)
- Stratum reticularosum (reticular layer)



Papillary layer

- **Dermal Papillae:** The dermal papillae contain capillaries (tiny blood vessels) and nerve endings, contributing to the nourishment and sensory perception of the epidermis. They are particularly abundant in areas of the skin with heightened sensitivity, such as the fingertips.
- **Blood Supply:** The capillaries within the dermal papillae provide oxygen and nutrients to the cells of the epidermis, supporting their metabolic needs.
- **Temperature Regulation:** The blood vessels within the papillary layer play a role in thermoregulation by helping to dissipate or retain heat as needed.
- **Sensory Reception:** The nerve endings in the papillary layer are involved in detecting various sensations, including touch, pressure, pain, and temperature. These sensory receptors contribute to our sense of touch and awareness of the environment.
- **Formation of Fingerprints:** The unique patterns of ridges and grooves formed by the dermal papillae are responsible for creating the distinctive patterns seen in fingerprints and footprints.
- **Support for Epidermis:** The papillary layer provides structural support for the epidermis and helps maintain the integrity of the skin.

Reticular layer of the dermis

- **Connective Tissue Matrix:** The reticular layer is primarily composed of dense irregular connective tissue. This tissue is made up of collagen and elastin fibers, which provide structural integrity, strength, and elasticity to the skin.
- **Strength and Resilience:** The dense arrangement of collagen fibers in the reticular layer gives the skin its strength and ability to resist stretching and mechanical forces. This layer helps maintain the shape and form of the skin.
- **Support for Blood Vessels and Glands:** Blood vessels, lymphatic vessels, and various accessory structures such as sweat glands and sebaceous glands are embedded within the reticular layer. These structures are important for thermoregulation, waste elimination, and secretion of various substances onto the skin's surface.
- **Nerve Endings:** The reticular layer contains sensory nerve endings that detect various sensations, including touch, pressure, pain, and temperature. These nerve endings play a crucial role in sensory perception and the body's response to external stimuli.
- **Collagen and Aging:** Collagen fibers in the reticular layer provide the skin's resilience and tautness. As collagen naturally breaks down over time due to aging and environmental factors, the skin can lose some of its elasticity and become prone to sagging and wrinkles.
- **Hair Follicles and Hair Muscles:** Hair follicles, which give rise to hair growth, are anchored within the reticular layer. Associated muscles (arrector pili muscles) contract to cause hair to stand erect, contributing to features like "goosebumps."

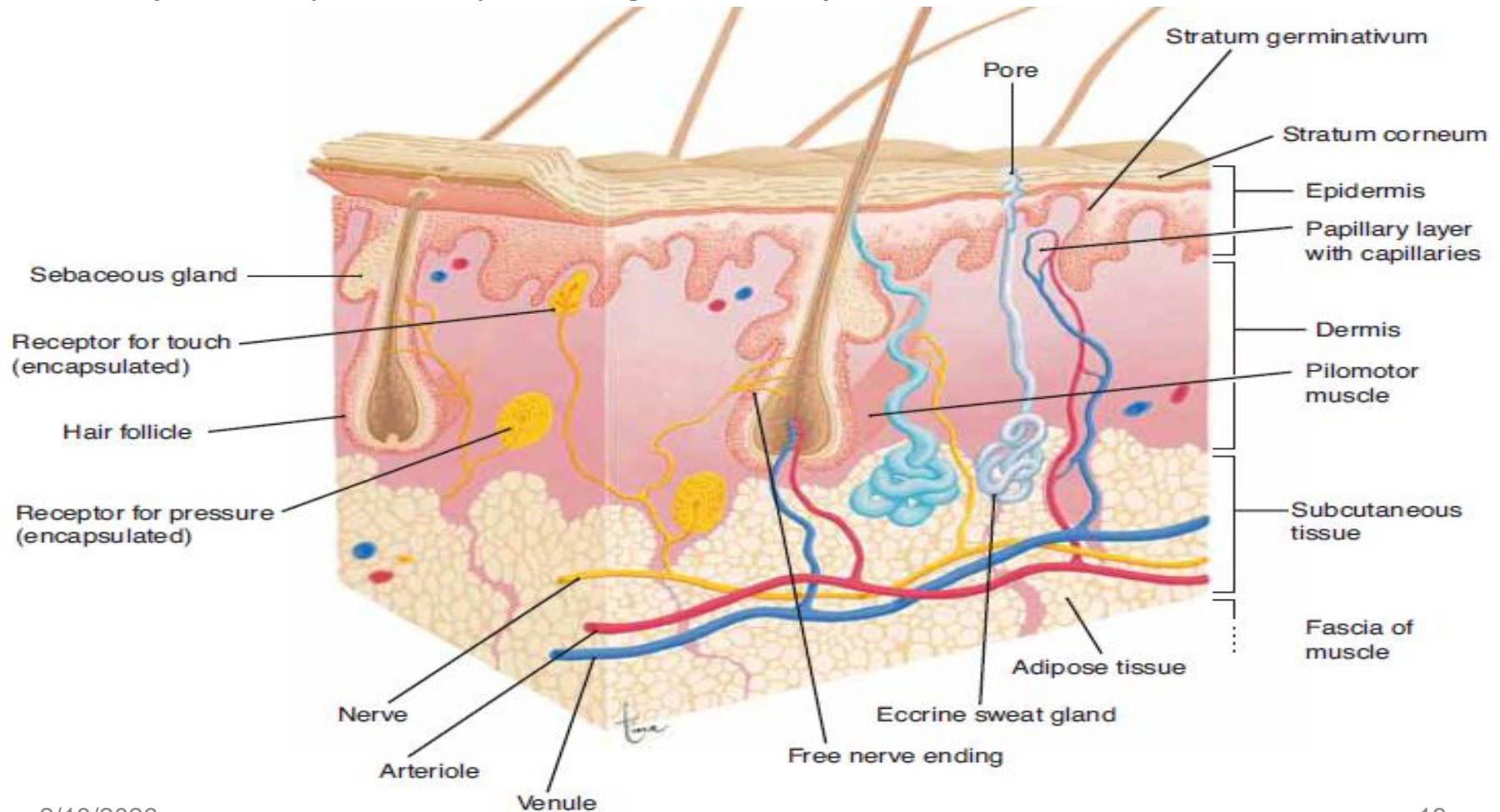
Nerve supply

- Free nerve ending
- Encapsulated nerve endings

Pancinian corpuscles; pressoreceptors. E.g. fingertips and CT

Meissner's corpuscles; tactile receptors. E.g. under the epidermis

Ruffini corpuscles; pain receptors. E.g. skin and joints



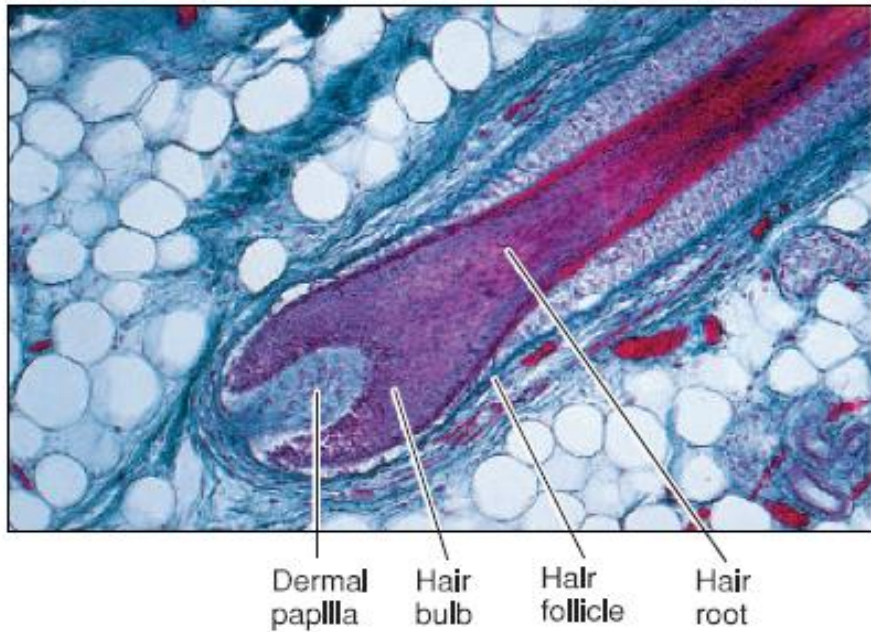
Free Nerve Endings: Free nerve endings are the simplest and most common type of sensory receptors in the skin and other tissues. They are unencapsulated nerve endings, meaning they lack a specialized structure around them. Free nerve endings are responsible for detecting a range of sensations, including pain, temperature, and itch.

- **Pain Receptors (Nociceptors):** Free nerve endings that detect pain are called nociceptors. They respond to potentially harmful stimuli, such as excessive pressure, temperature extremes, chemical irritants, or tissue damage. Nociceptors play a crucial role in alerting the body to potential threats and triggering pain responses.
- **Thermal Receptors:** Free nerve endings also include thermal receptors that respond to temperature changes. Some are sensitive to cold temperatures (cold receptors), while others respond to warmth (warm receptors).
- **Itch Receptors:** Free nerve endings are involved in the sensation of itching (pruritus). When certain substances come into contact with the skin, they can trigger the release of histamines and other compounds that activate itch receptors.

Encapsulated Nerve Endings: Encapsulated nerve endings are sensory receptors that are surrounded or encapsulated by specialized structures that enhance their sensitivity to specific stimuli. These encapsulated endings are more complex than free nerve endings and serve different sensory functions.

- **Pacinian Corpuscles:** These encapsulated receptors are sensitive to deep pressure and vibrations. They consist of layers of connective tissue that encase the nerve ending. Pacinian corpuscles are found in deeper layers of the skin, tendons, and some internal organs.
- **Meissner's Corpuscles:** Found in the dermal papillae of hairless skin, especially in fingertips, palms, and soles, Meissner's corpuscles are encapsulated receptors that respond to light touch and low-frequency vibrations. They are important for tactile discrimination and sensitivity.
- **Merkel Discs (Tactile Discs):** These encapsulated receptors are involved in light touch and tactile discrimination. They are found in the epidermis, especially in areas that require precise touch, such as the fingertips and lips.
- **Ruffini Corpuscles:** These receptors are found in the deep dermis and subcutaneous tissue. They respond to continuous pressure and skin distortion.

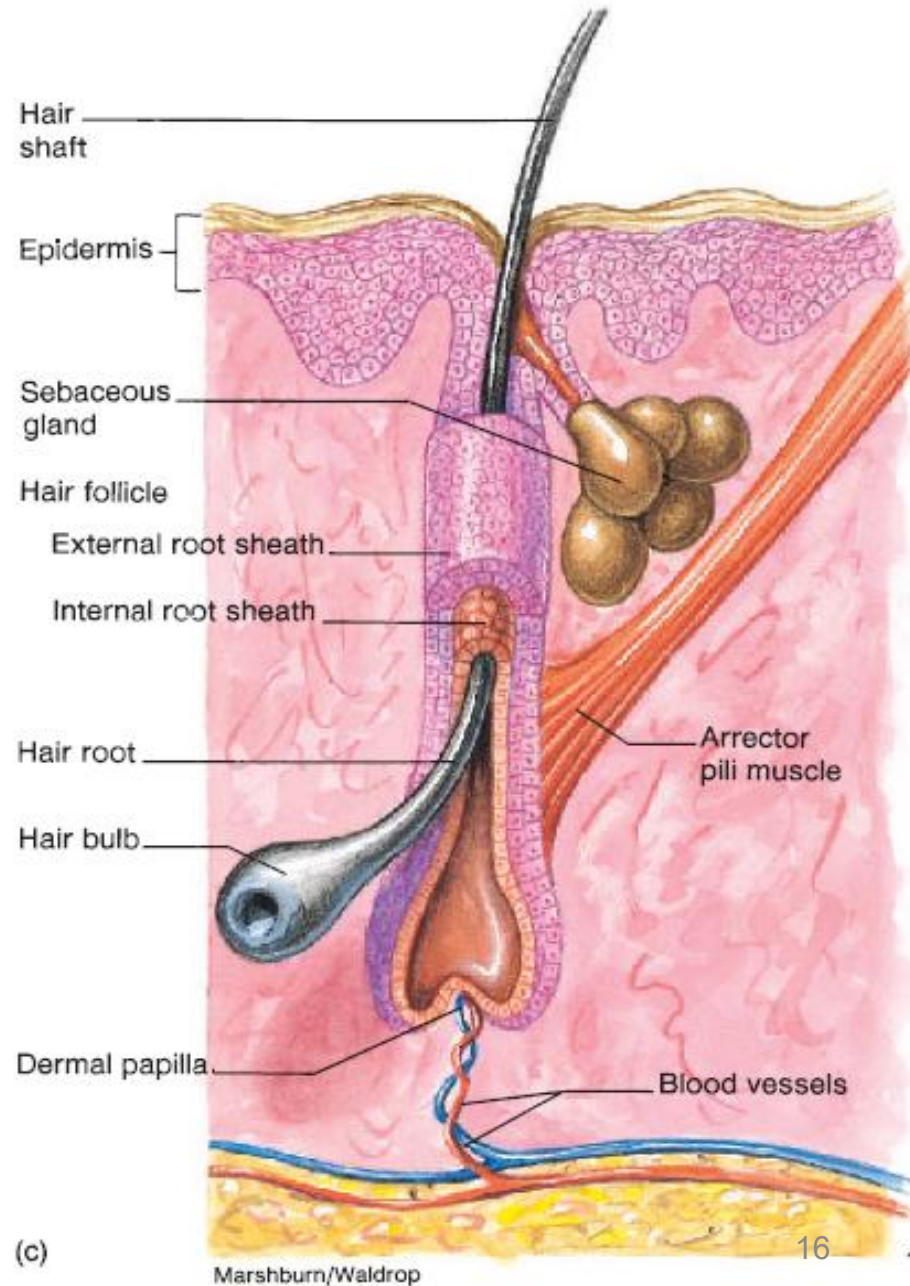
The structure of hair and the hair follicle



(a)

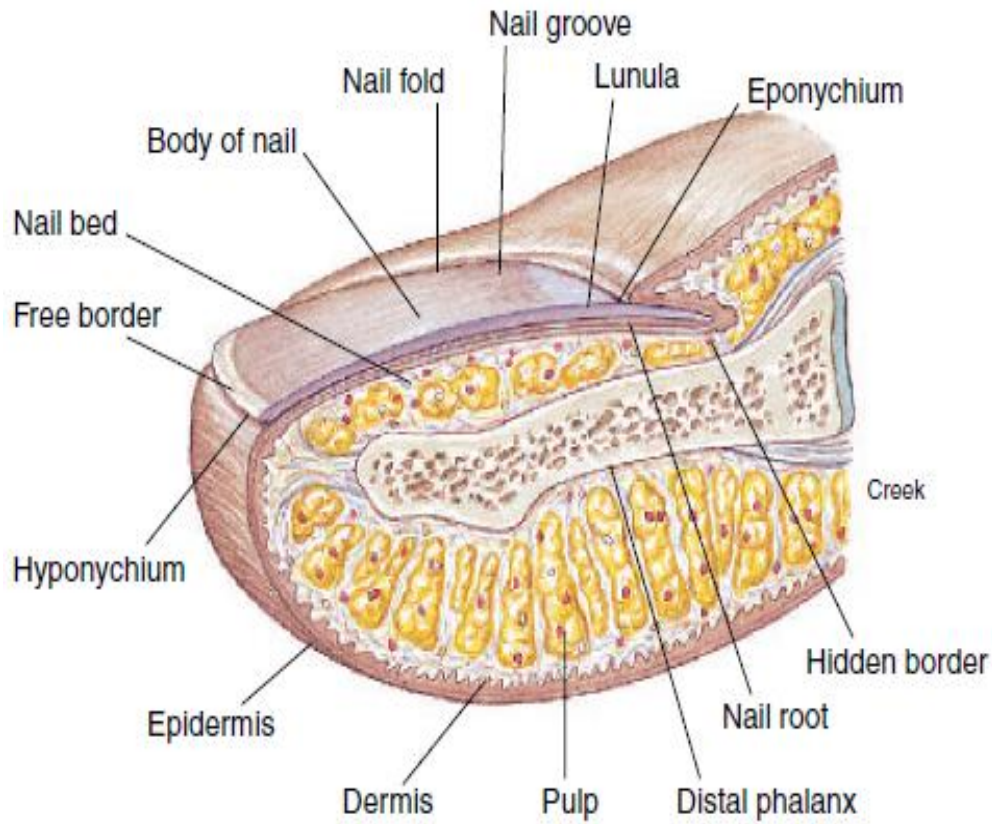


(b)

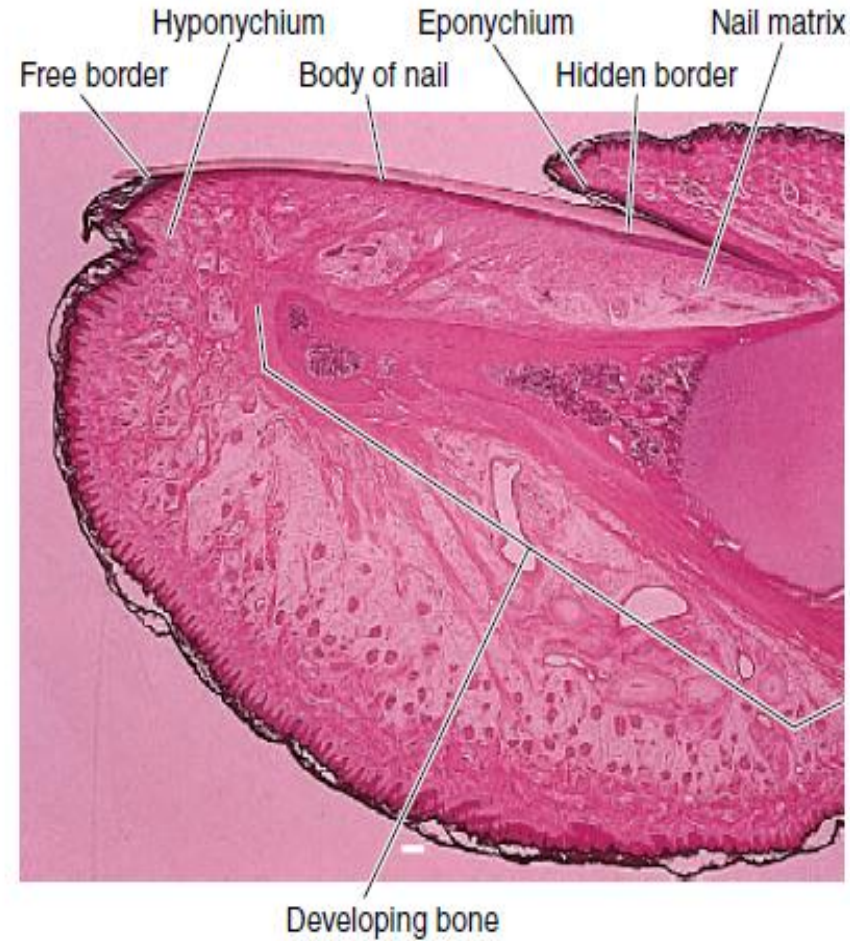


(c)

The fingertip and the associated structures of the nail

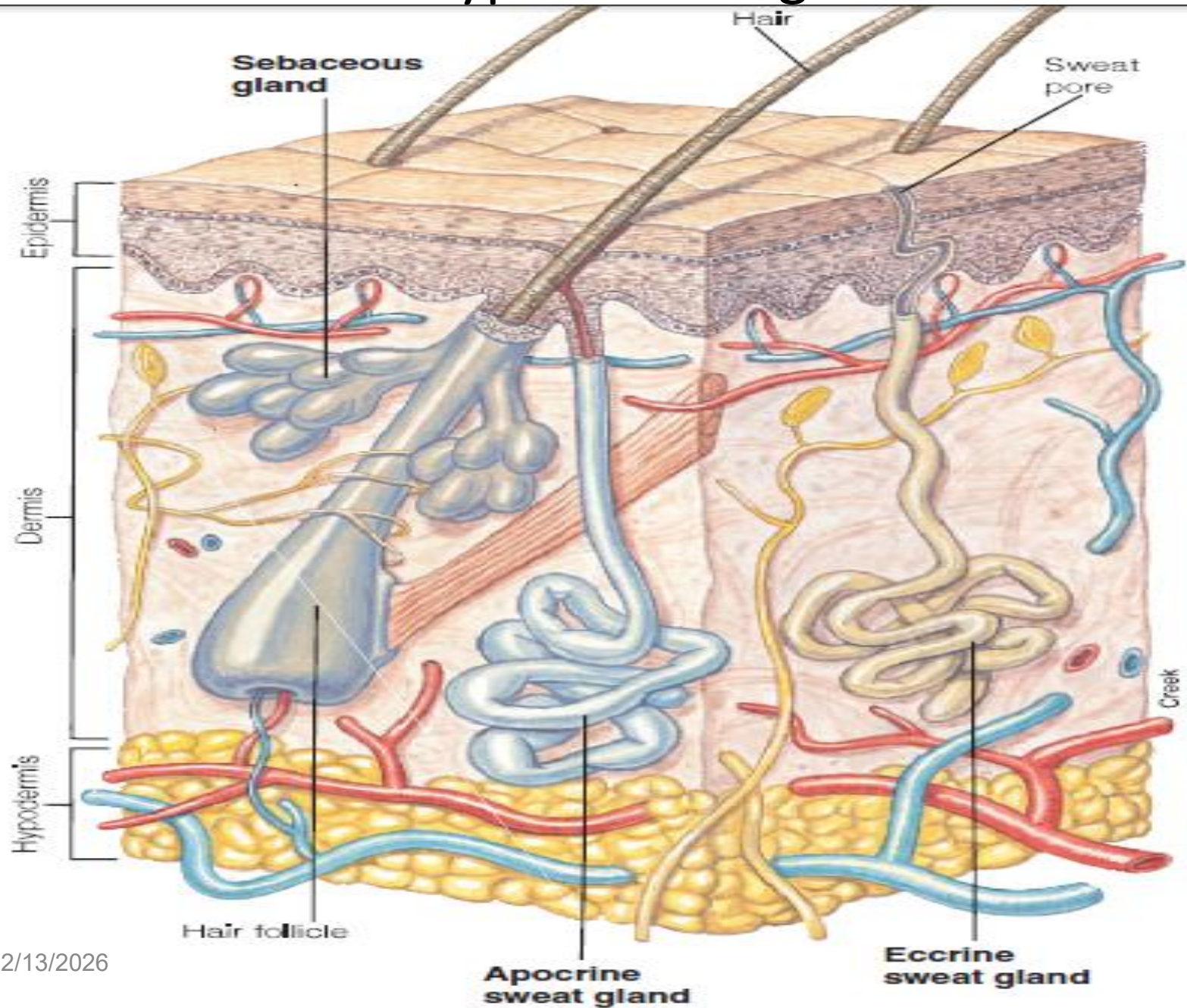


(a)



(b)

Types of skin glands.



Eccrine Sweat Glands: Eccrine sweat glands are the most numerous type of sweat glands and are found all over the body, with a higher concentration on the palms, soles, and forehead. They are involved in temperature regulation through a process known as thermoregulation.

Eccrine sweat glands secrete a clear, watery sweat directly onto the skin's surface. This sweat is primarily composed of water, electrolytes (such as sodium and chloride), and small amounts of other substances. When the body temperature rises, either due to physical activity, a warm environment, or fever, eccrine sweat glands are stimulated to produce sweat. As the sweat evaporates from the skin's surface, it helps cool the body down.

Apocrine Sweat Glands: Apocrine sweat glands are larger and found in specific areas of the body, such as the axillae (armpits), genital area, and areolae (around the nipples). Unlike eccrine sweat glands, apocrine sweat glands are not as directly involved in temperature regulation.

Apocrine sweat glands produce a thicker sweat that contains proteins and lipids. These glands are most active during emotional or stress-induced sweating, as well as during sexual arousal. The sweat produced by apocrine glands does not have a strong odor on its own, but when it comes into contact with bacteria on the skin's surface, it can lead to body odor.